

# Ruhvi — Project Tools & Integrations

Complete register of tools, integrations & add-ons | Generated 2026-08-24 | Next.js 15 + TypeScript + Supabase + Vercel

## 1. INFRASTRUCTURE, HOSTING & BUSINESS SERVICES

| TOOL / SERVICE | CATEGORY        | PURPOSE                                                                                     | STATUS |
|----------------|-----------------|---------------------------------------------------------------------------------------------|--------|
| Hostinger      | Domain          | Domain purchase/registration for ruhvi.in.                                                  | Active |
| Cloudflare     | DNS & Security  | DNS hosting/proxy for ruhvi.in subdomains; also provides Turnstile bot protection in-app.   | Active |
| Vercel         | Hosting         | Production website hosting & deployments; Vercel Speed Insights for performance monitoring. | Active |
| GitHub         | Version Control | Source code repository & version control; CI/CD via Vercel git integration.                 | Active |
| PostHog        | Analytics / CX  | Customer behavior analysis: product analytics, funnels, session replay, heatmaps.           | Active |
| Zoho           | Email           | Business email service on custom domain (support@ruhvi.in, MX records).                     | Active |
| Airtel SIM     | Mobile / SMS    | Mobile connectivity & SMS (OTP delivery, WhatsApp Business messaging).                      | Active |

## 2. DATABASE & BACKEND

| TOOL                        | PURPOSE                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supabase (PostgreSQL + RLS) | Primary database: users, products, orders, carts, audit_logs, order_events, AI state, support tickets, wallets. SDKs: @supabase/supabase-js, @supabase/ssr. |
| Upstash Redis               | Serverless Redis for caching (cacheGet/Set/Delete/Wrap) and fixed/sliding-window rate limiting via Lua scripts. SDK: @upstash/redis.                        |

## 3. AUTHENTICATION

| TOOL          | PURPOSE                                                                           |
|---------------|-----------------------------------------------------------------------------------|
| Firebase Auth | Customer identity: email/password, Google, Facebook, phone/OTP, provider linking. |

|                                      |                                                                                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Firestore Admin (custom REST + jose) | Server-side user ops (lookup, password-reset links, guest creation) via Google Identity Toolkit — avoids firebase-admin ESM bundler conflicts. |
| Custom JWT Session Bridge (jose)     | Verifies Firebase ID tokens via Google JWKS, mints Supabase-signed __session cookie for RLS identity.                                          |
| RBAC & Support Auth                  | DB-driven role permissions (role_permissions, wildcard module.*) for portal access.                                                            |

#### 4. AI GATEWAY (CUSTOM MULTI-PROVIDER ENGINE)

| PROVIDER                              | PURPOSE                                      |
|---------------------------------------|----------------------------------------------|
| Google Gemini (@google/generative-ai) | Primary LLM — product/SEO content + chatbot. |
| OpenAI                                | GPT fallback provider (REST).                |
| Anthropic Claude                      | Claude fallback provider (REST).             |
| OpenRouter                            | Aggregated model fallback (REST).            |
| DeepSeek                              | DeepSeek fallback provider (REST).           |
| Custom Gateway                        | Pluggable custom endpoint provider.          |

Engine features: credential rotation with cooldown, model health tracking, error classification, PII redaction, failover, diagnostics — in `src/lib/ai/`.

#### 5. EMAIL & MESSAGING

| TOOL                                  | PURPOSE                                                                                                                     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Resend                                | All transactional email (welcome, order status, payment, support tickets, password reset) via Handlebars templates.         |
| Brevo / Sendinblue                    | Marketing email: abandoned-cart, win-back, birthday/anniversary campaigns; contact management; campaign stats.              |
| Brevo MCP (@modelcontextprotocol/sdk) | SSE MCP server exposing Brevo actions as AI tools (send_transactional_email, create_or_update_contact, get_campaign_stats). |
| WhatsApp Cloud API (Meta)             | Template-based order/shipping notifications + webhook with hub.challenge verification.                                      |

#### 6. PAYMENTS & SHIPPING

| TOOL | PURPOSE |
|------|---------|
|------|---------|

|            |                                                                                                    |
|------------|----------------------------------------------------------------------------------------------------|
| PhonePe PG | UPI payment gateway (SHA-256 + salt X-VERIFY checksum, PAY_PAGE), UAT + prod + simulated fallback. |
| Shiprocket | Logistics: order creation, courier/AWB assignment, webhook status updates.                         |

## 7. NOTIFICATIONS & ANALYTICS

| TOOL                         | PURPOSE                                                                                 |
|------------------------------|-----------------------------------------------------------------------------------------|
| Firebase Cloud Messaging     | Web push via VAPID + service worker (firebase-messaging-sw.js).                         |
| OneSignal                    | Cross-platform push notifications; admin sends via REST API.                            |
| Meta Pixel + Conversions API | Client fbq + server-side Graph API mirror with eventID dedupe.                          |
| Google Analytics 4           | Pageview/event/ecommerce tracking (gtag).                                               |
| PostHog                      | Product analytics & customer behavior (posthog-node server client + posthog-js client). |
| Vercel Speed Insights        | Real-user performance monitoring (Web Vitals).                                          |

## 8. MEDIA, SECURITY & ERROR MONITORING

| TOOL                    | PURPOSE                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------|
| Cloudinary              | Image/video/PDF upload (unsigned preset) + URL transformations for product zoom/thumbnails. |
| Cloudflare Turnstile    | Checkout bot protection, server-verified.                                                   |
| Sentry (@sentry/nextjs) | Error monitoring for client, server & edge runtimes; source map upload; org: ruhvi-jewels.  |
| Capacitor               | Native iOS/Android wrapper (app in.ruhvi.app, webDir out).                                  |

## 9. CLIENT-SIDE UI & 3D LIBRARIES

| LIBRARY                               | PURPOSE                                             |
|---------------------------------------|-----------------------------------------------------|
| three.js / React Three Fiber / Drei   | Interactive 3D hero ring scene.                     |
| PhotoSwipe (react-photoswipe-gallery) | Fullscreen zoomable product image galleries.        |
| Recharts                              | Admin & support dashboard charts / AI analytics.    |
| React Hook Form + Zod                 | Form handling & validation (checkout, auth, admin). |

|                       |                       |
|-----------------------|-----------------------|
| Lucide React          | UI icon set.          |
| react-hot-toast       | Toast notifications.  |
| clsx + tailwind-merge | Class name utilities. |

Ruhvi Jewels (ruhvi.in) — Project Tools & Integrations Register | For internal reference only. Keep this document safe.